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# Thumb to Moon: how wide it looks is not how wide it is

A thumb held up at the Moon shows how wide the Moon looks. It does not show how many kilometers across the Moon is.

To estimate that width, use how wide the Moon looks, and a distance found another way.

Suppose a 1 centimeter gap, held 110 centimeters from one eye, just covers the Moon. This is a rounded teaching example, not a measurement you have made.

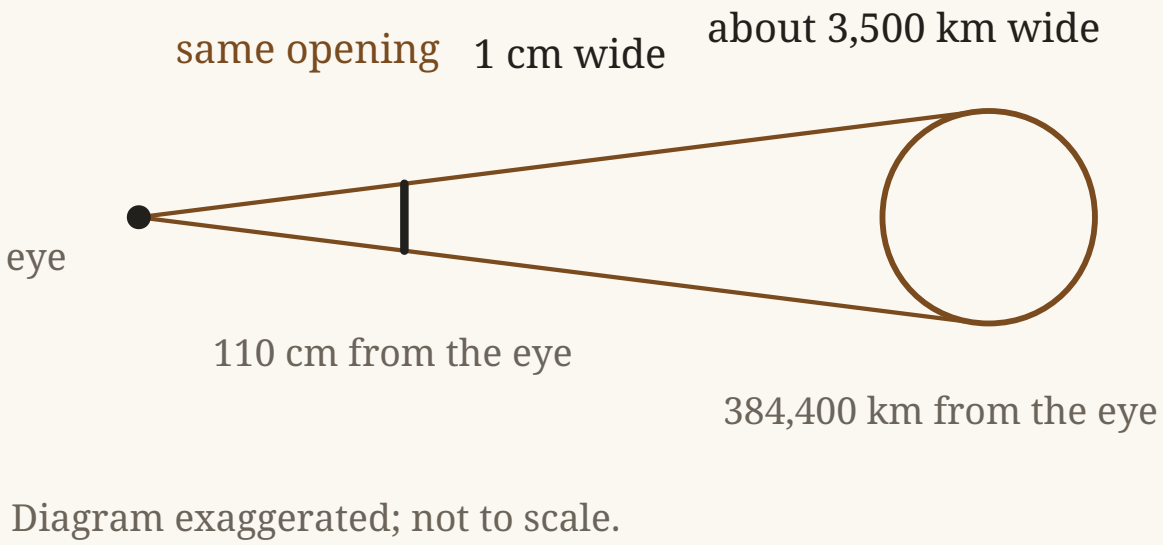


Diagram exaggerated; not to scale. A 1 cm gap at 110 cm and the Moon share the drawn opening.

The gap's distance is 110 times its width. The Moon fills the same view, so the Moon's distance is also about 110 times its width.

NASA's average distance to the Moon is 384,400 kilometers. An average is not the distance on one particular night.

384,400 divided by 110 is about 3,495 kilometers.

So the Moon is about 3,500 kilometers across.

Check afterward. NASA's Moon facts give a radius of about 1,740 kilometers, the line from the center to the edge. The diameter, the width through the middle, is twice the radius, about 3,480 kilometers. That is close to this estimate.

A thumb's width, and its distance from your eye, vary. A thumb alone is not a calibrated ruler. Do not look at the Sun this way.

## Sources

- NASA Moon facts: average distance 384,400 kilometers, radius about 1,740 kilometers.

## Key facts

- A 1 centimeter gap at 110 centimeters is a rounded teaching example, not a measurement you made.
- The Moon is then about 110 times farther away than it is wide.
- 384,400 divided by 110 is about 3,495 kilometers, so about 3,500 kilometers across.
- After that estimate, NASA's radius of about 1,740 kilometers gives a diameter of about 3,480 kilometers.

The printed page is Thumb to Moon (PDF).